

# Noida Institute of Engineering and Technology, Greater Noida

(An Autonomous Institute)

## School of Computer Science and Engineering

### PROJECT PROGRESS REPORT CARD

Session: 2026 - 2027

Project ID / Group No. :- \_\_\_\_\_

| Roll No.  | Student Name       | Section | Project Title                                                                                                       | Supervisor Name       | Grade (A/B/C) |
|-----------|--------------------|---------|---------------------------------------------------------------------------------------------------------------------|-----------------------|---------------|
| <Roll No> | Aryan Raj Singh    | <Sec>   | AMASE: A Harness-Ablation Framework for Measuring Cost-Quality Trade-offs in Autonomous Software Engineering Agents | Mr. <Supervisor Name> |               |
| <Roll No> | <Team Member Name> | <Sec>   |                                                                                                                     |                       |               |
| <Roll No> | <Team Member Name> | <Sec>   |                                                                                                                     |                       |               |
| <Roll No> | <Team Member Name> | <Sec>   |                                                                                                                     |                       |               |

| Meeting No. | Meeting Date & Time | Work Assigned by Supervisor                                                                                                                      | Supervisor's Remarks | Supervisor's Signature |
|-------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------|
| 1           |                     | Phase 0 — Finalise problem statement, literature survey and scope; set up repository, Python/uv toolchain and version control.                   |                      |                        |
| 2           |                     | Phase 0 — Implement the Docker execution sandbox: per-task provisioning, stdout/stderr capture, teardown, and turn/time/cost ceilings.           |                      |                        |
| 3           |                     | Phase 0 — Build the minimal ReAct core on LangGraph with the seven tools; integrate the Terminal-Bench 2.x adapter; resolve one task end to end. |                      |                        |
| 4           |                     | Phase 1 — Instrument all model and tool calls to the OpenTelemetry GenAI semantic conventions; stand up self-hosted Langfuse; verify traces.     |                      |                        |
| 5           |                     | Phase 1 — Implement cost accounting and the experiment runner (config matrix, n repetitions, mean and spread); record the baseline.              |                      |                        |

|    |  |                                                                                                                                                   |  |  |
|----|--|---------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| 6  |  | Phase 1 — Build the DeepEval CI gate in GitHub Actions; demonstrate that a regressing pull request is blocked automatically.                      |  |  |
| 7  |  | Phase 2 — Implement and measure arms A1 (environment context) and A2 (pre-completion checklist) at $n \geq 3$ against baseline.                   |  |  |
| 8  |  | Phase 2 — Implement and measure arms A3 (doom-loop detection), A8 (reasoning budget) and A4 (compaction strategy); run the interaction factorial. |  |  |
| 9  |  | Phase 3 — Implement arm A6 (hybrid retrieval vs agentic search) and arm A7 (static model routing); produce the cost-quality curve.                |  |  |
| 10 |  | Phase 4 — Author the held-out decontaminated task set; run confirmatory experiments; prepare the dashboard, final report and research paper.      |  |  |

Signature of Supervisor: \_\_\_\_\_  
 \_\_\_\_\_

Signature of HOD / Project Coordinator: